



BSc (Hons) in Information Technology
Year 4 Semester 2

IT4100 – SQA

Lab Exercise 03 – Cyclomatic Complexity

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Question 1

What is CC?

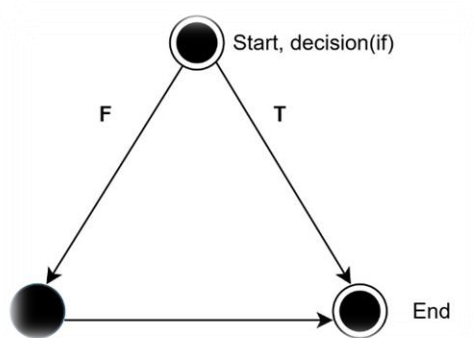
Code Coverage (CC) is a software testing metric that measures the extent to which the source code is exercised during software testing. Code coverage analysis is used to detect code that is not exercised during testing, assess the quality of tests, and enhance software reliability. Code coverage is widely used in automated testing and CI/CD pipelines, but high code coverage does not necessarily mean bug-free software.

Its Usage.

Code Coverage is employed to uncover areas of the code that are not yet tested, assess the efficiency of test cases, enhance code quality, minimize bugs, and ensure that the critical logic of code is tested before its release. Code Coverage also enables QA and development teams to track the progress of testing and ensure quality in automated testing.

Question 2

```
public void recQuickSort (int left, int right)
```



CC value

$$V(G) = e - n + 2$$

$$= 3 - 3 + 2$$

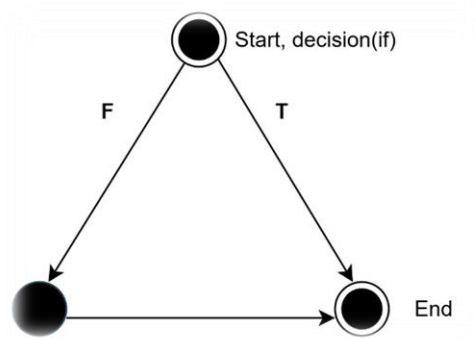
$$= \underline{\underline{2}}$$

$$V(G) = d + 1$$

$$= 1 + 1$$

$$= \underline{\underline{2}}$$

public void setCurrentValue (float val)



CC value

$$V(G) = e - n + 2$$

$$= 3 - 3 + 2$$

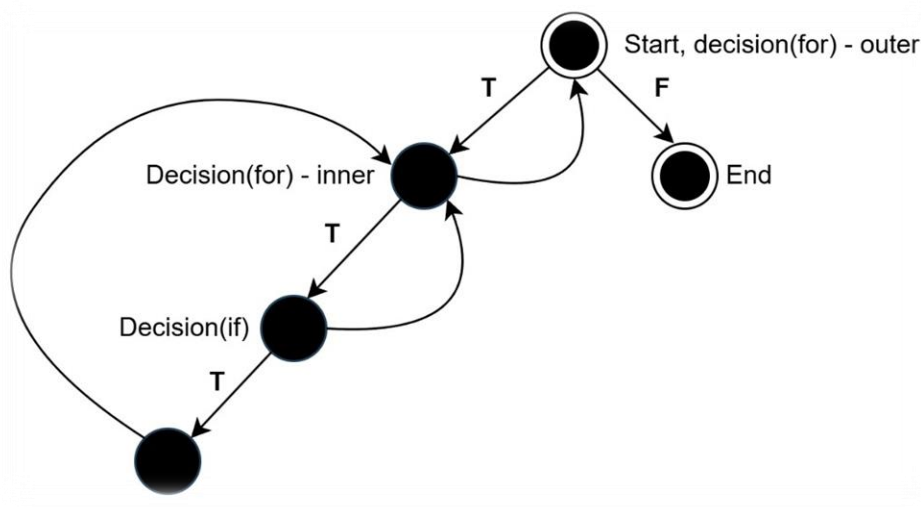
$$= \underline{\underline{2}}$$

$$V(G) = d + 1$$

$$= 1 + 1$$

$$= \underline{\underline{2}}$$

```
public void bubbleSort ()
```



CC value

$$V(G) = e - n + 2$$

$$= 7 - 5 + 2$$

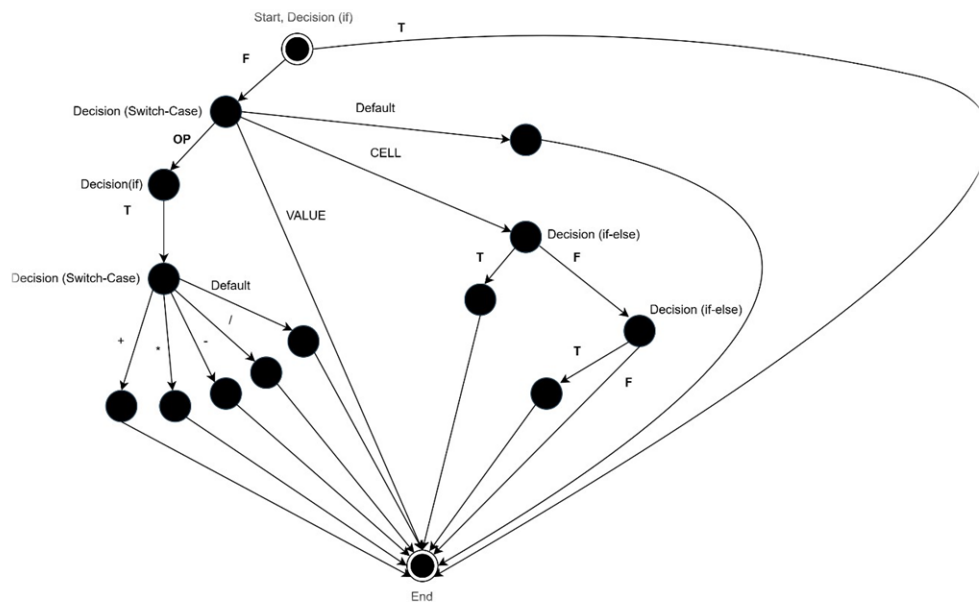
$$= \underline{\underline{4}}$$

$$V(G) = d + 1$$

$$= 3 + 1$$

$$= \underline{\underline{4}}$$

public float evaluateFormula (Node n)



CC value

$$\begin{aligned}
 V(G) &= e - n + 2 \\
 &= 24 - 15 + 2 \\
 &= \underline{\underline{11}}
 \end{aligned}$$

$$\begin{aligned}
 V(G) &= d + 1 \\
 &= 5 + 1 \\
 &= \underline{\underline{6}}
 \end{aligned}$$

Question 03

```
public void recQuickSort(int, int);
Code:
  0: iload_2
  1: iload_1
  2: isub
  3: ifgt          7
  6: return
  7: aload_0
  8: getfield      #2          // Field theArray:[D
 11: iload_2
 12: daload
 13: dstore_3
 14: aload_0
 15: iload_1
 16: iload_2
 17: dload_3
 18: invokevirtual #11          // Method partitionIt:(IID)I
 21: istore        5
 23: aload_0
 24: iload_1
 25: iload        5
 27: iconst_1
 28: isub
 29: invokevirtual #10          // Method recQuickSort:(II)V
 32: aload_0
 33: iload        5
 35: iconst_1
 36: iadd
 37: iload_2
 38: invokevirtual #10          // Method recQuickSort:(II)V
 41: return
```

public void recQuickSort(int left, int right)

$$\begin{aligned} V(G) &= d + 1 \\ &= 1 + 1 \\ &= \underline{\underline{2}} \end{aligned}$$

```

public void setCurrentValue(float);
Code:
  0: aload_0
  1: getfield      #9          // Field selectedRow:I
  4: iconst_m1
  5: if_icmpeq     16
  8: aload_0
  9: getfield      #10         // Field selectedColumn:I
 12: iconst_m1
 13: if_icmpne     17
 16: return
 17: aload_0
 18: getfield      #32         // Field cells:[[LCell;
 21: aload_0
 22: getfield      #9          // Field selectedRow:I
 25: aaload
 26: aload_0
 27: getfield      #10         // Field selectedColumn:I
 30: aaload
 31: fload_1
 32: invokevirtual #96         // Method Cell.setValue:(F)V
 35: aload_0
 36: invokevirtual #53         // Method repaint:()V
 39: return

```

public void setCurrentValue (float val)

$$\begin{aligned}
 V(G) &= d + 1 \\
 &= 2 + 1 \\
 &= \underline{\underline{3}}
 \end{aligned}$$


```

public void bubbleSort();
Code:
  0: aload_0
  1: getfield      #3              // Field nElements:I
  4: iconst_1
  5: isub
  6: istore_1
  7: iload_1
  8: iconst_1
  9: if_icmple     57
12: iconst_0
13: istore_2
14: iload_2
15: iload_1
16: if_icmpge     51
19: aload_0
20: getfield      #2              // Field a:[D
23: iload_2
24: daload
25: aload_0
26: getfield      #2              // Field a:[D
29: iload_2
30: iconst_1
31: iadd
32: daload
33: dcmpl
34: ifle          45
37: aload_0
38: iload_2
39: iload_2
40: iconst_1
41: iadd
42: invokevirtual #9              // Method swap:(II)V
45: iinc          2, 1
48: goto          14
51: iinc          1, -1
54: goto          7
57: return
}

```

public void bubbleSort()

$$\begin{aligned}
 V(G) &= d + 1 \\
 &= 3 + 1 \\
 &= \underline{\underline{4}}
 \end{aligned}$$

```

public float evaluateFormula(Node);
Code:
  0: fconst_0
  1: fstore_2
  2: aload_1
  3: ifnonnull      8
  6: fload_2
  7: freturn
  8: aload_1
  9: getfield      #83          // Field Node.type:I
 12: tableswitch   { // 0 to 2
                0: 40
                1: 148
                2: 153
                default: 214
            }
 40: aload_0
 41: aload_1
 42: getfield      #84          // Field Node.left:LNode;
 45: invokevirtual #59          // Method evaluateFormula:(LNode;)F
 48: fstore_2
 49: aload_1
 50: getfield      #85          // Field Node.op:C
 53: tableswitch   { // 42 to 47
                42: 106
                43: 92
                44: 145
                45: 120
                46: 145
                47: 134
                default: 145
            }
 92: fload_2
 93: aload_0
 94: aload_1
 95: getfield      #86          // Field Node.right:LNode;
 98: invokevirtual #59          // Method evaluateFormula:(LNode;)F
101: fadd
102: fstore_2
103: goto         145
106: fload_2
107: aload_0

```

public float evaluateFormula(Node n)

$$\begin{aligned}
 V(G) &= d + 1 \\
 &= 5 + 1 \\
 &= \underline{\underline{6}}
 \end{aligned}$$

	Source Code	Byte Code
public void recQuickSort (int left, int right)	2	2
public void setCurrentValue (float val)	2	3
public void bubbleSort ()	4	4
public float evaluateFormula (Node n)	11	6

Question 04

Explain why public void setCurrentValue(float val) and public float evaluateFormula (Node n) methods are reporting different values for source and byte codes.

1. Explanation for the CC value difference of the public void setCurrentValue (float val)

Since the 'if' condition being a compound condition, there are 2 separate decision statements in the byte code. Hence, the value reported by the byte code is higher than the value reported by the source code.

2. Explanation for the CC value difference of the public float evaluateFormula (Node n)

Usually, a decision node has a maximum of two outgoing edges for the true and false paths. However, a decision node which represent a switch condition has more than two outgoing edges. Hence, in such situations, the values reported by $e - n + 2$ and $d + 1$ are different from each other. In fact, the correct cyclomatic complexity (CC) value is reported by $e - n + 2$

--End of the Report--